Name:______________

Date:______________

Logarithmic Functions & Models

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can convert between logarithmic and exponential forms.**

RELATED SKILLS

- I — Convert between logarithmic and exponential forms.
- I — Interpret a logarithm as an exponent.
- I — Read logarithmic notation and identify the base, argument, and corresponding exponential statement.

☐ **2. I can evaluate a logarithm using its exponential meaning.**

RELATED SKILLS

- I — Rewrite exponential expressions using a common base when possible.
- D — Interpret a logarithm as an exponent.

☐ **3. I can expand or condense an expression using product, quotient, and power properties.**

RELATED SKILLS

- I — Apply the logarithm power property in either direction.
- I — Apply the logarithm product property in either direction.
- I — Apply the logarithm quotient property in either direction.
- A — Simplify an algebraic expression using operation properties.

☐ **4. I can solve an exponential equation using logarithms.**

RELATED SKILLS

- I — Isolate an exponential expression and apply a logarithm.
- A — Apply the logarithm power property in either direction.
- A — Isolate a specified variable in an equation or formula.

☐ **5. I can solve a logarithmic equation using exponential form and check domain restrictions.**

RELATED SKILLS

- I — Combine or isolate logarithms before converting to exponential form.
- I — Reject solutions that make a logarithm argument nonpositive.
- A — Convert between logarithmic and exponential forms.

☐ **6. I can calculate and interpret doubling time or half-life.**

RELATED SKILLS

- I — Translate a doubling-time or half-life question into an exponential equation.
- D — Explain a model parameter in the language of its context.
- A — Attach and interpret units for rates, inputs, and outputs.
- A — Isolate an exponential expression and apply a logarithm.

7. I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

RELATED SKILLS

- I — Identify vertical asymptotic behavior of a logarithmic graph.
- D — Connect parameters in a formula to features of its graph.
- D — Determine domain restrictions from denominators, even roots, and logarithms.
- A — Reject solutions that make a logarithm argument nonpositive.

8. I can explain how exponential and logarithmic functions undo one another.

RELATED SKILLS

- D — Interpret a logarithm as an exponent.
- A — Exchange input and output values when interpreting an inverse.
- A — Verify inverse functions using composition.

9. I can solve a multi-step exponential or logarithmic equation.

RELATED SKILLS

- D — Combine or isolate logarithms before converting to exponential form.
- D — Isolate an exponential expression and apply a logarithm.
- A — Apply the logarithm power property in either direction.
- A — Apply the logarithm product property in either direction.
- A — Apply the logarithm quotient property in either direction.
- A — Reject solutions that make a logarithm argument nonpositive.

10. I can determine an intersection of exponential functions graphically or algebraically.

RELATED SKILLS

- D — Isolate an exponential expression and apply a logarithm.
- A — Check an ordered pair in both equations of a system.
- A — Compare exponential growth rates expressed in different forms.
- A — Locate and interpret the intersection of two lines.

11. I can rewrite a discrete exponential model as an equivalent continuous exponential model.

RELATED SKILLS

- D — Convert between equivalent discrete- and continuous-growth forms.
- A — Convert a percent increase or decrease to a growth or decay factor.
- A — Isolate an exponential expression and apply a logarithm.

12. I can solve and interpret an exponential or logarithmic modeling problem.

RELATED SKILLS

- D — Explain a model parameter in the language of its context.
- D — Identify initial amount, rate, and time interval in context.
- A — Attach and interpret units for rates, inputs, and outputs.
- A — Combine or isolate logarithms before converting to exponential form.
- A — Isolate an exponential expression and apply a logarithm.
- A — Restrict a model to meaningful contextual inputs.